

February 23, 2026

Mr. Thomas Keane, MD, MBA
Assistant Secretary for Technology Policy
Office of the National Coordinator for Health Information Technology
Attention: HHS Health Section AI RFI
330 C Street S.W.
Washington DC 20201

Re: HHS Health Sector AI RFI

Dr. Mr. Keane,

Aidoc is grateful for the opportunity to respond to the Request for Information: *Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care*.

The need for clinical AI is immense. It was recently published that an estimated 900,000 diagnostic errors occur in the US annually, leading to nearly 400,000 deaths and serious disablements.¹ The need for AI further increases with the increasing shortage of clinicians, significantly reducing access to care, especially in rural communities.

We believe clinical AI provides the largest step function in care delivery in recent history. Many of those diagnostic errors, delays and access gaps are preventable, paving the way for the largest public health benefit in decades.

At the same, we have the largest technological shift with clinical AI finally reaching diagnostic-level accuracy. Prior clinical AI technologies were valuable yet narrow in scope, focusing on one disease at a time. Only in the last few months foundation model-based AI advancements have proven themselves in clinical settings, unlocking the ultimate potential impact of identifying every disease throughout a patient's scan, and doing so with very high accuracy, paving the way to the transformative opportunity mentioned above. However, this is not without addressing and mitigating its new risks, and establishing the required changes to regulatory framework.

As a pioneer in clinical AI, Aidoc has moved beyond the era of fragmented "point solutions" to establish the industry's most robust health system infrastructure for scalable, responsible AI deployment. Our proprietary AI Operating System (aiOS™) serves as the critical clinical layer bridging the gap between medical imaging and the Electronic Medical Record (EMR). By orchestrating the flow of data and AI insights, we transform passive practices into proactive clinical action across six major service lines (including neurology, cardiovascular, and oncology), demonstrating that AI can augment the standard of care without disrupting existing workflows.

¹ Newman-Toker DE; Just how many diagnostic errors and harms are out there, really? It depends on how you count;BMJ Quality & Safety 2025;34:355-360.

Aidoc's commitment to clinical rigor and technical innovation is reflected in the clinical value we deliver and the scale at which our solutions are deployed across both regulated and non-regulated domains.

- **Global Reach & Scale:**

Aidoc is currently deployed across more than 250 health systems consisting of 1,600 hospitals in 44 U.S. states and 19 countries worldwide. This extensive footprint provides us with direct insight into the diverse operational and clinical challenges faced by healthcare providers across different systems and geographies.

- **Vast Clinical Intelligence:**

To date, aiOS™ has analyzed more than 100 million patient cases, leveraging the largest portfolio of FDA-cleared AI devices in its category. This scale provides a unique longitudinal perspective on how clinical AI can impact patient outcomes, workflow efficiency, and population health management.

aiOS™ enables data normalization, continuous performance monitoring, and built-in governance controls. It allows health systems to deploy and manage multi-condition clinical AI solutions without requiring infrastructure re-architecture. With more than 100 million patient cases analyzed, aiOS™ is the most widely deployed clinical AI platform in healthcare.

- **Frontier Foundation Models:**

Developed in collaboration with global AI industry leaders AWS and NVIDIA, CARE™ (Clinical AI Reasoning Engine) is Aidoc's proprietary diagnostic foundation model. CARE enables the rapid development of multi-output comprehensive medical devices with high levels of accuracy, allowing for significant enhancement of both clinical utility and operational efficiency.

In January 2026, Aidoc received FDA clearance for the first multi-triage application powered by CARE, encompassing double-digit indications within a single abdominal CT scan and covering numerous acute abdominopelvic imaging-detectable conditions. The milestone followed a Breakthrough Device designation from the FDA, reflecting the clinical relevance and the need for the device in U.S healthcare. This application represents the first of many CARE-powered solutions currently under development at Aidoc.

The mission driving this technology is the mitigation of diagnostic harm. As mentioned above, each year, an estimated 900,000 Americans are impacted by diagnostic errors resulting in death or permanent disability and the vast majority occurring in hospital settings. Clinical AI has the documented ability to help mitigate this harm. For example, studies show that AI-enabled

technology can yield a 40% reduction in time to treatment² and 2-day reduction in Intensive Care Unit length of stay.³

We believe AI represents an evolutionary step-change in the practice of medicine. We welcome the opportunity to partner with HHS to establish a progressive, scalable policy approach that accelerates adoption while upholding the highest bar for patient safety and consumer trust. By aligning regulation, reimbursement, and R&D investments, we can reduce the uncertainty that impedes innovation and ensure AI is deployed to enhance productivity, reduce clinician burden, and improve health outcomes for communities across America.

Aidoc's response to HHS RFI

Aidoc's experience demonstrates that AI has moved beyond fragmented "point solutions" to become a critical infrastructure layer for modern medicine. To fully realize the clinical potential of AI, we encourage HHS to embrace a forward-looking policy approach that supports patient safety, diagnostic quality, and sustained innovation. Establishing clear standards for clinical AI governance will help ensure reliable performance while enabling continued technological advancement.

Accordingly, this response focuses on key insights derived from Aidoc's large-scale deployment experience and outlines a strategic roadmap centered on:

Regulatory and Standards Evolution

The FDA has demonstrated increasing openness to regulatory innovation. We recommend continued adaptation of premarket pathways and the development of post-market standards that reflect the unique characteristics and development pace of AI technologies. Such evolution should support a safe, effective, and efficient quality framework for AI-enabled clinical solutions, regardless of whether they are classified as medical or non-medical devices.

Payment Innovation

Establishing reimbursement pathways for AI-driven improvements in diagnostic quality could help unlock broader clinical and population-level benefits. Innovative payment models may support adoption of high-impact technologies that improve outcomes, enhance access, and reduce downstream healthcare costs.

By aligning policy development with clinical implementation realities, HHS can help ensure that technological innovation is deployed safely, effectively, and at scale for the benefit of both clinicians and patients.

² Chalian, et al. "Enhancing Pulmonary Embolism Management: The Impact of AI-Based Triage and Prioritization on Diagnostic Timeliness and Treatment Initiation." RSNA Annual Meeting, 2024.

³Burch et al. Improving Patient Outcomes with an AI-Enhanced Pulmonary Embolism Response Team in a Large Healthcare Network - 2024 PERT Consortium. Available at:<https://pertconsortium.org/wp-content/uploads/2024/09/Improving-Patient-Outcomes-with-an-AI-Enhanced-Pulmonary-Embolism-Response-Team-in-a-Large-Healthcare-Network.pdf>

Please note that Aidoc chose to address questions where it believes it can provide meaningful insight to HHS and did not respond to every question detailed in the RFI.

Responses to Individual Questions:

HHS question 1: What are the biggest barriers to private sector innovation for AI for healthcare and its adoption and use in clinical care?

Aidoc believes that the most significant barriers to private sector innovation in healthcare AI - and to its responsible adoption in clinical care - are structural rather than technological. While AI capabilities continue to advance rapidly, the surrounding healthcare ecosystem has not evolved at the same pace. Reimbursement models, validation standards, implementation processes, and workforce preparedness remain fragmented.

These structural gaps create uncertainty for developers, misaligned incentives for health systems, friction during deployment, and inconsistent governance across the AI lifecycle. Collectively, they slow innovation, limit investment, and delay patient access to clinically meaningful technologies.

The primary barriers fall into these interconnected categories:

- Potential regulatory opportunities and lifecycle governance gaps
- Misaligned reimbursement and financial incentives
- Fragmented evaluation and monitoring infrastructure
- Workforce readiness, trust, and liability concerns

Addressing these systemic challenges in a coordinated manner would accelerate responsible innovation while strengthening patient safety, trust, equity, and long-term sustainability of AI-enabled clinical care.

i. Potential Regulatory Opportunities and Lifecycle Governance Gaps

i.a Full AI Lifecycle Quality Governance

Innovation is constrained by the lack of harmonized federal governance across the full AI Total Product Lifecycle (TPLC).

The current regulatory landscape is imbalanced: some AI tools are deployed with minimal lifecycle oversight, while others face unclear or very robust requirements. This creates market uncertainty—allowing some products to enter clinical use without defined standards while potentially slowing innovation for others.

Aidoc believes foundational AI Quality Management System (QMS) principles should apply across the AI lifecycle, including clear definition of intended use and users, instructions for use, required user training, defined validation, post-

deployment surveillance, change management, and performance transparency & explainability.

Establishing baseline AI lifecycle governance expectations would reduce uncertainty, clarify liability and shared accountability, and most important - support patient safety, similar to foundational controls applied to low-risk medical devices (such as class I devices, which require a QMS, but a premarket review by FDA is not needed). We believe that unified basic standardization would support medical and non medical devices, and will also help a unified federal standard motion, allowing to reduce state legislation and fragmentation in this domain.

i.b Risk–Benefit Frameworks That Better Capture System-Level Clinical Value

Current regulatory benefit assessments often focus primarily on diagnostic performance metrics, particularly in novel regulatory pathways (De Novo submissions). However, AI-enabled clinical tools frequently generate value through workflow optimization and system-level impacts that can influence patient outcomes.

These efficiency-related impacts are often treated as operational rather than quality outcomes, limiting their influence on regulatory risk–benefit determinations. A more comprehensive framework incorporating system-level clinical impact would better reflect real-world AI value.

i.c Alignment of Validation Pathways With Clinical Risk and Evidence Maturity

Validation requirements remain inconsistent across device categories. In particular, they can impose disproportionate burdens on multi-indication and multimodal AI systems that are essential for scalable clinical decision support.

We appreciate that FDA has increasingly adopted innovative regulatory approaches, and we strongly support the continued use and expansion of such pathways. Further alignment between validation rigor, clinical risk, technological maturity, and real-world evidence generation would reduce unnecessary development friction while preserving high standards of safety and effectiveness.

ii. Misaligned Reimbursement and Financial Incentives

Current payment policies do not adequately incentivize improvements in front-end diagnostic quality and preventive care.

The absence of clear reimbursement pathways for AI-enabled detection, triage, and care coordination technologies limits health system investment in solutions that can prevent

high-harm diagnostic errors, reduce downstream treatment costs, and support health equity.

Without aligned financial incentives, adoption decisions are often driven by short-term operational return on investment rather than long-term system value.

This misalignment slows deployment of technologies that may generate substantial population-level clinical and economic benefits but do not fit neatly within existing reimbursement structures.

iii. Fragmented Evaluation and Continuous Monitoring Infrastructure

There remains fragmentation between initial regulatory clearance (when applicable, for medical devices) and sustained real-world performance evaluation.

Developers and healthcare organizations lack standardized and widely adopted frameworks for holistic AI evaluation that extend beyond traditional performance metrics.

Meaningful assessment should include:

- Patient outcome impact
- Workflow integration and operational effectiveness
- Longitudinal performance stability
- Detection and mitigation of model drift
- Equity and population health considerations

Fragmented evaluation structures lead to duplicated effort, slow knowledge sharing, and reduced collective confidence in AI-enabled clinical systems.

A coordinated post-market evidence generation and monitoring infrastructure, which can be established as part of an AI Total Product Lifecycle (TPLC) framework as mentioned in item i.a above, would promote transparency, accelerate learning across healthcare systems, and support continuous safety and effectiveness assurance.

v. Workforce Preparedness, Trust, and Liability Considerations

Workforce readiness is a critical component of safe AI adoption.

Most practicing clinicians receive limited formal training in AI literacy, which may reduce confidence in interpreting AI outputs and exercising appropriate clinical oversight, in addition to increasing the likelihood of usage-related risks, first and foremost, the risk of automation bias.

Expanding AI literacy within clinical education and professional development programs would support safer, more effective integration of AI technologies into clinical workflows.

Liability and accountability uncertainty further constrain adoption. Healthcare systems often perceive regulatory and legal risk during early deployment phases, particularly when governance boundaries between developer, institution, and clinician are not clearly defined.

Clarified accountability frameworks and potential safe harbor mechanisms could reduce adoption hesitation, especially for smaller or resource-constrained institutions such as rural healthcare providers.

HHS question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why?

We believe that HHS has a tremendous opportunity to adapt regulatory and payment policies to meaningfully accelerate the impact of AI on clinical care.

i. Regulatory Policy:

ii.a Establish Harmonized AI Lifecycle Governance Across Clinical Settings

A central barrier to effective AI adoption is the lack of federally harmonized governance across the full AI TPLC. Importantly, this gap does not operate in isolation - it directly contributes to fragmentation in evaluation infrastructure and to uncertainty experienced by the clinical workforce.

HHS should prioritize establishing baseline lifecycle governance expectations for clinical AI tools, regardless of whether they qualify as medical devices or undergo premarket review. Foundational Quality Management System (QMS) expectations should include:

- Clear definition of intended use and intended users
- Transparent and actionable instructions for use
- Adequate clinical validation and performance characterization
- Post-deployment surveillance and safety monitoring
- Structured change management for model updates
- Appropriate explainability and transparency to end users

Without consistent baseline governance, safety and quality uncertainty increases, and interpretive burden is shifted to healthcare institutions. In practice, health systems are forced to build ad hoc oversight, monitoring, and validation structures—often duplicating efforts and applying inconsistent standards. Standardized lifecycle expectations would clarify shared accountability between developers and healthcare systems, reduce redundancy, and create the structural foundation upon which coordinated evaluation systems and workforce confidence can be built.

We would like to further zoom in on two interrelated aspects that are inherent components of a comprehensive TPLC framework: (1) fragmented evaluation and continuous monitoring infrastructure, and (2) workforce preparedness, trust, and liability considerations. These are not separate challenges, but downstream manifestations of incomplete lifecycle governance. Addressing them explicitly within the TPLC structure would significantly strengthen safe and scalable AI adoption.

Fragmented Evaluation and Continuous Monitoring Infrastructure

The absence of harmonized lifecycle governance contributes directly to fragmentation between initial regulatory clearance (when applicable for medical devices) and sustained real-world performance evaluation.

Where baseline expectations for post-deployment monitoring, performance reassessment, and change management are not clearly defined at the federal level, developers and healthcare organizations lack standardized and widely adopted frameworks for holistic AI evaluation. As a result, evaluation efforts often remain siloed—focused narrowly on premarket performance metrics rather than continuous, real-world effectiveness.

Meaningful assessment should include:

- Patient outcome impact
- Workflow integration and operational effectiveness
- Longitudinal performance stability
- Detection and mitigation of model drift
- Equity and population health considerations

Fragmented evaluation structures lead to duplicated efforts, slower knowledge sharing, and reduced collective confidence in AI-enabled clinical systems. Critically, they also weaken the feedback loop that should connect post-market evidence generation back into lifecycle governance and model improvement.

A coordinated post-market evidence generation and monitoring infrastructure—established as part of a comprehensive AI TPLC framework—would operationalize the governance principles described above. By embedding structured surveillance, shared data standards, and continuous learning mechanisms into the lifecycle itself, such an infrastructure would promote transparency, accelerate cross-system learning, and support ongoing safety and effectiveness assurance.

Workforce Preparedness, Trust, and Liability Considerations

Lifecycle governance and monitoring fragmentation also have direct implications for clinician trust and liability perception.

When governance expectations are unclear and post-market monitoring is inconsistent, clinicians are left uncertain about the reliability, scope, and limitations of AI tools deployed in their workflows.

This uncertainty compounds existing gaps in AI literacy and increases the risk of inappropriate reliance or automation bias.

Workforce readiness is therefore inseparable from lifecycle governance. Clear intended use definitions, transparent validation standards, explainability requirements, and structured change management processes all support clinicians in understanding when and how AI outputs should inform decision-making.

At present, most clinicians receive limited formal training in AI literacy, which may reduce confidence in interpreting AI outputs and exercising appropriate oversight. Expanding AI literacy within undergraduate medical education, residency programs, and continuing professional development initiatives would strengthen safe integration—but these efforts must be supported by consistent governance signals and reliable monitoring infrastructure.

Liability and accountability uncertainty further constrain adoption. Healthcare systems often perceive heightened regulatory and legal risk during early deployment phases, particularly when governance boundaries between developer, institution, and clinician are not clearly defined. In the absence of harmonized lifecycle standards, responsibility is diffused and often shifted downstream.

Clarified accountability frameworks tied to adherence to defined lifecycle governance standards—would reduce hesitation and align incentives. Such clarity is especially important for smaller or resource-constrained institutions, including rural healthcare providers, which may lack the capacity to independently construct robust AI governance ecosystems

ii.b Modernize Risk–Benefit Assessment Frameworks

Current regulatory benefit frameworks often focus narrowly on diagnostic accuracy metrics, particularly in novel pathways such as De Novo classifications. However, AI-enabled clinical tools frequently generate value through workflow optimization and system-level improvements that influence patient outcomes.

For example, AI that accelerates identification and discharge of low-risk patients in crowded emergency departments can reduce bed shortages, improve triage efficiency, and enable faster care for critically ill patients. These impacts materially influence morbidity and mortality but are often categorized as “efficiency” rather than “quality,” limiting their regulatory weight in risk–benefit determinations.

HHS should encourage regulatory approaches that:

- Recognize workflow acceleration and throughput improvements as clinical quality drivers
- Incorporate system-level outcomes into benefit assessment
- Evaluate downstream impact on time-to-treatment, diagnostic delay, and resource allocation

A broader risk–benefit framework would better reflect real-world clinical value and reduce disincentives for meaningful innovation.

ii.c Align Validation Pathways With Clinical Risk and Real-World Evidence

It is evident the FDA has been working tirelessly to innovate safe and effective methods of product evaluation. We'd like to encourage that continued motion. Certain radiology software products continue to rely heavily on pre market studies. While we believe in the importance of robust pre market assessment, there are ways to balance the load better between pre and post market, relying more heavily on real world evidence.

These requirements can disproportionately burden multi-indication and multimodal AI systems that are essential for scalable clinical decision support.

HHS should prioritize:

- Expanded acceptance of real-world evidence in validation
- Alignment of evidentiary rigor with clinical risk and technological maturity
- Flexible validation pathways for multi-indication and adaptive AI systems

Reducing unnecessary development friction—while preserving high safety standards—would accelerate responsible deployment of clinically meaningful AI technologies.

ii. Payment Policy:

CMS has taken important steps to create progressive reimbursement programs that can spur technological innovation. The current ACCESS and TEMPO are key starting point initiatives that may meaningfully catalyze collaboration between health systems and industry technology partners to address care gaps and accelerate innovation. However, the current programs are focused primarily on chronic conditions and ambulatory care.

CMS has an opportunity to extend this impact further to address the crisis of diagnostic error and substantially mitigate serious harm. Today, an estimated 900,000 serious diagnostic errors occur in the US annually, leading to nearly 400,000 deaths and serious disablements.⁴ These errors are largely preventable, with nearly 90% of them occurring in a hospital setting (inpatient or ED). By creating reimbursement pathways based on improving measurable front-end diagnostic quality for high-harm conditions, CMS can elevate Diagnostic Harm to become a policy priority and meaningfully reduce preventable harm related to diagnostic errors.

Current reimbursement pathways are slow to evolve, narrowly scoped compared to the foundational infrastructure AI represents, and often poorly suited to AI-enabled care, particularly

⁴ Newman-Toker DE; Just how many diagnostic errors and harms are out there, really? It depends on how you count;BMJ Quality & Safety 2025;34:355-360.

when payment models require a direct physician work component rather than recognizing automated diagnostic and coordination value.

One could argue that certain AI-enabled activities can be financed under existing fee-for-service mechanisms. However, relying on fee-for-service alone creates structural barriers to adoption and limits the public health impact of AI:

1. It weakens incentives to invest in earlier detection interventions where the return accrues over years, such as cardiovascular disease or cancer, because the entity funding the intervention is often not the entity capturing downstream savings.
2. This dynamic disproportionately disadvantages community hospitals that care for higher proportions of Medicare and Medicaid beneficiaries and operate on thinner margins, even though these institutions could realize outsized benefit from improved diagnostic quality and proactive care pathways.
3. Fee-for-service does not consistently reward solutions that reduce total cost of care, which reduces incentives for both developers and health systems to prioritize AI that lowers avoidable utilization, prevents complications, and improves longitudinal outcomes.

As an example, heart disease is the leading cause of death in the United States, yet proven tools to support population screening - such as coronary artery calcium (CAC) scoring - remain underused due largely to reimbursement gaps. Although nearly half of U.S. adults have measurable coronary calcium and elevated CAC scores increase coronary event risk sevenfold, most insurers do not routinely cover CAC analysis. As a result, access is limited and follow-up is inconsistent, with approximately 30% of moderate-to-severe cases receiving no preventive therapy. This creates a significant missed opportunity for early, equitable intervention before costly and life-threatening cardiac events occur.

AI-enabled CAC analysis fundamentally changes the reimbursement equation by leveraging imaging that is already being performed. Rather than requiring new tests, AI can retrospectively and prospectively analyze existing chest CT scans - performed on millions of Americans each year - to identify incidental coronary calcium at scale. When paired with AI-driven care management workflows that ensure appropriate follow-up, this approach transforms CAC screening from an underutilized diagnostic tool into a scalable preventive care pathway. Evidence shows opportunistic screening significantly increases statin initiation, demonstrating clear downstream clinical action and measurable value.

For CMS and other payers, reimbursing AI-based CAC analysis and management represents a high-impact, infrastructure-light investment aligned with value-based care and health equity goals. The cost of inaction is substantial - missed incidental CAC in the Medicare chest CT population alone represents billions of dollars in first-year medical costs and lost life-years. By establishing reimbursement pathways for AI-driven CAC analysis and follow-up management, for example, CMS can incentivize health systems to close a major preventive cardiology gap, reduce avoidable hospitalizations, and shift cardiovascular spending from emergency response to proactive risk reduction.

The CAC example provided, is just an example, intended to illustrate that traditional code-based reimbursement may not fully address these needs, but thoughtful innovation in payment design could unlock adoption at national scale and deliver population-level impact unmatched by any prior diagnostic reform.

Coupled with this opportunity, we recognize the complexity of modernizing payment policy. We propose two potential paths forward:

First, CMS could modernize and expand existing payment pathways for emerging technologies, particularly in hospital-based and emergency care settings where AI can meaningfully reduce diagnostic delay and prevent harm. CMS could consider establishing pathways that allow certain AI technologies to be recognized as distinct, separately payable services, rather than being bundled into existing procedure or imaging codes or tied to physician work. Many AI systems perform clinically meaningful, resource-intensive functions, such as independent data analysis, risk stratification, and clinical decision support, that are not adequately reflected in current bundled payments. Creating mechanisms to unbundle and appropriately value these technological services would help ensure payment accurately reflects the resources involved, support sustained innovation, and promote access to technologies that improve care quality, efficiency, and outcomes.

Second, as an alternative to purely code-based reimbursement, CMS should consider establishing a dedicated Diagnostic Quality Improvement payment pathway, potentially through CMMI demonstration authority, that rewards measurable reductions in diagnostic delay, preventable harm, and avoidable downstream utilization for high-impact conditions. Such a model could be structured around performance-based payments tied to validated outcome metrics, enabling health systems to adopt AI-enabled infrastructure that demonstrably improves front-end diagnostic accuracy and care coordination. Thoughtful innovation in payment design would unlock adoption at national scale and deliver population-level impact unmatched by prior diagnostic reforms.

HHS question 3: For non-medical devices, what novel legal and implementation issues exist, and what role should HHS play?

The use of AI in clinical care outside traditional medical device regulatory pathways introduces novel legal, governance, and implementation challenges. These challenges primarily stem from structural gaps in accountability architecture rather than from the intrinsic technological risk of AI itself.

As clinical AI systems increasingly influence clinical workflows and decision-making processes, policy focus should shift toward establishing baseline trust, safety, and responsibility frameworks while preserving flexibility for innovation. In particular, clarifying governance expectations across the AI lifecycle would help address many of the uncertainties associated with AI tools that fall outside conventional medical device regulation.

We believe that if the recommendation outlined in Section II.A—establishing harmonized AI lifecycle governance across clinical settings—is implemented, many of the legal and implementation challenges associated with non-medical-device clinical AI applications could be substantially mitigated. A consistent lifecycle governance framework would help define accountability boundaries, support transparent validation and monitoring practices, and reduce ambiguity regarding the roles of developers, healthcare institutions, and clinicians.

Accordingly, HHS can play a critical role in advancing baseline governance standards, promoting workforce readiness, and facilitating shared evidence and monitoring infrastructures that support safe and responsible deployment of clinical AI across regulatory categories.

HHS question 5: *How can HHS best support private-sector activities (e.g., accreditation, certification, credentialing) to promote AI use?*

We believe that thoughtful evolution of payment and reimbursement design could play a critical role in enabling the national-scale adoption of clinical AI and supporting population-level clinical and operational impact that cannot be achieved through technology advancement alone. Aligning economic incentives with demonstrated clinical value and real-world performance may help facilitate the responsible integration of AI into clinical care pathways.

In parallel, we believe that greater harmonization and standardization across the TPLC for clinical AI developers could strengthen confidence in AI-enabled systems, thereby supporting broader private-sector adoption and clinical reliance on validated technologies. Both of these dimensions are further discussed in our response to Question 2 above.

HHS question 6: *Where has AI in clinical care met or exceeded expectations, and where has it fallen short? What novel tools hold the most promise?*

AI tools in clinical care have most consistently met or exceeded performance and expectations in well-defined, workflow-integrated use cases, particularly imaging triage, early detection of high-risk conditions, and care coordination, where they have improved timeliness of diagnosis and

treatment^{5,6,7}, reduced ED⁸, inpatient^{9,10}, and ICU length of stay^{11,12} and improved follow-up and intervention rates.^{13,14}

Beyond the published literature, adoption trends reinforce this progress. On the Aidoc platform alone, the average number of use cases deployed per health system has grown from roughly three just a few years ago to more than a dozen today. Clinicians and administrators are expanding deployment because they are seeing tangible clinical and operational value.

By contrast, AI has fallen short when there is misalignment between an algorithm's intended use and real world clinical workflows, or when financial incentives do not support the required changes in practice. Even strong technical performance does not translate into clinical impact if ground truth definitions are inconsistent, integration is weak, or adoption is superficial. A clear example is incidental findings. As discussed in the payment reform section, many health systems choose not to implement the workflow changes required to act on these findings because the financial incentives are unclear or misaligned.

Adoption is also uneven. Community based systems, particularly those operating primarily in fee for service environments, often struggle to identify sustainable payment pathways. As a result, the benefits of AI are not being distributed uniformly across the healthcare system.

As a specific example: as AI technology continues to make rapid advancement, more comprehensive detection becomes possible at scale. Broad adoption of, for example, comprehensive imaging AI, would enable significant improvements to diagnostic speed and accuracy, particularly in acute care settings.

⁵ Radiology AI - Wiklund, Peder, and Koshlar Medson. "Use of a Deep Learning Algorithm for Detection and Triage of Cancer-associated Incidental Pulmonary Embolism." Proceedings of the Radiological Society of North America (RSNA) 2023 Annual Meeting, 18 Oct. 2023, <https://doi.org/10.1148/ryai.220286>.

⁶ Chalian, et al. "Enhancing Pulmonary Embolism Management: The Impact of AI-Based Triage and Prioritization on Diagnostic Timeliness and Treatment Initiation." RSNA Annual Meeting, 2024.

⁷ Chokhawala, H., H. Cuellar, R. Shah, and A. Savardekar. "Enhancing Stroke Care Efficiency through AI Vendor Transition: A Comparative Study of Workflow Metrics Before and After Implementation." SNIS Conference, 2024, LSUHS, Shreveport, LA. Abstract presentation.

⁸ Davis MA, Rao B, Cedeno PA, Saha A, Zohrabian VM. Machine Learning and Improved Quality Metrics in Acute Intracranial Hemorrhage by Noncontrast Computed Tomography. *Curr Probl Diagn Radiol*. 2022;51(4):556-561. doi:10.1067/j.cpradiol.2020.10.007

⁹ Petry, Michael et al. "Decreased Hospital Length of Stay for ICH and PE after Adoption of an Artificial Intelligence-Augmented Radiological Worklist Triage System." *Radiology research and practice* vol. 2022 2141839. 18 Aug. 2022, doi:10.1155/2022/2141839

¹⁰ Davis, Melissa A., et al. "Machine Learning and Improved Quality Metrics in Acute Intracranial Hemorrhage by Noncontrast Computed Tomography." *Current Problems in Diagnostic Radiology*, vol. 51, no. 4, July–August 2022, pp. 556-561, Elsevier, <https://doi.org/10.1067/j.cpradiol.2020.10.007>.

¹¹ Improving Patient Outcomes with an AI-Enhanced Pulmonary Embolism Response Team in a Large Healthcare Network - Burch et al. 2024 PERT Consortium

¹² Gupta, Kavish, MD, et al. "Mechanical Thrombectomy, Artificial Intelligence and the Activation of a Pulmonary Embolism Response Team." PERT Consortium 2023, abstract presentation.

¹³ Kostiuk, Valentyna, et al. "Artificial Intelligence Improves Initial Evaluation and Follow-Up of Patients with AAA." Abstract, Vascular Annual Meeting, 2024. Yale School of Medicine, New Haven, CT.

¹⁴ Randazzo, Michael, et al. "Hospital-Wide Integration of a Natural Language Processing Algorithm to Detect Inferior Vena Cava Filters in Imaging Reports and Improve Device Removal Rates." ACC Abstracts, 2024.

Looking ahead, AI tools with the greatest potential are those that support proactive and longitudinal care management, including timely, accurate identification of high-risk patients; ensuring appropriate follow-up; improving care coordination; and enabling earlier intervention to improve outcomes and reduce avoidable costs.

HHS question 7: Which decision makers drive AI adoption within healthcare organizations, and what are the administrative hurdles?

Aidoc's experience shows that when a broad enough portfolio provides real clinical value, AI adoption can shift from departmental "point solutions" to enterprise-wide strategy.

Such effective adoption requires a synchronized triple sponsorship model and a governance structure that treats AI as a high powered always-on-assistant rather than just software.

i. Influential Roles and Decision-Makers

The adoption of clinical AI is driven by a multi-disciplinary trifecta of leadership that ensures the technology is sharpening clinical judgment rather than replacing it:

- Triple Sponsorship: Successful integration requires three distinct sponsors:
 - Executive Sponsor: Owns the enterprise impact and provides the authority to allocate resources and remove obstacles.
 - Clinical Sponsor: A physician champion (e.g., CMO, CMIO or Service Line Leader) who provides direction on workflows and ensures the tool is designed for the "joy of practice" and improvement of patient care.
 - IT/Technical Partner: Anticipates integration needs and owns the long-term reliability and security of the system.
- Clinical Champions : These front-line clinicians are essential for generating excitement and surfacing early challenges. They act as microphones to build conviction across the medical staff.
- Enterprise AI Governance Committees: Many health systems are formalizing AI Councils that include legal, ethics, and clinical stakeholders. Their role is to be the Chief Ethical Officers of the organization's data, treating algorithmic bias with the same urgency as a hospital-acquired infection.

ii. Primary Administrative Hurdles

It has been our experience and observation that even with strong clinical support, several administrative bottlenecks frequently stall implementation:

- The Integration Gap & Change Fatigue: The greatest hurdle is dropping a technically sound solution into a poorly understood clinical reality. If AI creates more noise or cognitive load, adoption fails. Success requires mapping the end-to-end patient journey to ensure the AI becomes an invisible collaborator.
- Vendor Credentialing Fatigue: Redundant security and compliance reviews currently take

6–12 months per facility. Aidoc addresses this through models like our Implementation Palooza, an on-site working session that brings all stakeholders (clinical, IT, and vendors) into one room to troubleshoot technical roadblocks in real-time.

- Governance Wait Times: Traditional governance often slows down routine changes. Effective governance must be agile and enable speed with control and utilizing technology for continuous monitoring so oversight is an ongoing practice rather than a one-time approval.
- Lack of ROI framework: In today's financial environment, AI adoption requires a clear return on investment. The challenge is that clinical AI often delivers value downstream and across silos. Earlier detection can reduce ICU length of stay, prevent complications, and decompress the ED. These are enterprise level gains, but they rarely sit neatly within one department's budget. Lastly, oftentimes incentives are insufficient for improving longer term care.
- Health systems that evaluate AI only on short term billing lift risk undervaluing it. The right lens is enterprise ROI, including reduced length of stay, avoided adverse events, improved throughput, performance under value based contracts, and workforce stability.
- Measurement of the Human Dividend: Organizations often struggle to move beyond traditional ROI (like billing efficiency) to measure the human dividend such as the reduction in burnout, concerns about what urgent findings may be waiting in a queue of unread images, or the return of joy to clinical practice.

By supporting standardized certification and interoperable monitoring infrastructure, HHS can help reduce these administrative frictions. This allows healthcare leaders to focus on the clinical AI renaissance and reclaim the art of medicine by letting AI handle the science of data while humans focus on the connection with the patient.

HHS question 9: What challenges do patients and caregivers want AI to address, and what concerns do they hold?

Leveraging AI to improve diagnostic quality is a key focus of patients and caregivers. Diagnostic error harm represents a significant but often under-recognized challenge in American healthcare.

Imaging AI can play a critical role in diagnostic workflows within emergency and inpatient settings and is central to the diagnosis of several high-impact conditions that contribute substantially to national error rates, including stroke, pneumonia, pulmonary embolism/deep vein thrombosis, and lung cancer. Research suggests that many inpatient diagnostic errors are preventable, with nearly 85% of identified adverse outcomes potentially avoidable. Strengthening hospital-based diagnostic workflows therefore represents a high-leverage opportunity to address the broader diagnostic safety challenge.

Beyond safety, AI also offers an unprecedented opportunity to improve access to high-quality care across diverse and underserved populations. Clinical AI systems can be deployed across care environments- from large integrated delivery networks to remote rural hospitals. At a time when healthcare systems are under pressure to expand access, AI can help level disparities by

enabling more consistent delivery of high-quality clinical decision support regardless of geographic or institutional setting.

Federal initiatives focused on rural health transformation present an important opportunity to align policy with emerging evidence. Care delays are not inevitable; coordinated policy, technology deployment, and infrastructure development can reduce preventable harm and improve outcomes. Ultimately, while a patient's ZIP code may determine which hospital they enter, it should not determine the sophistication of the technology used to support life-saving clinical decisions.

Nevertheless, concerns regarding AI adoption remain, primarily around quality, effectiveness, and safety. We believe these concerns can be meaningfully addressed through the establishment of harmonized AI TPLC governance frameworks, as discussed in response to Question 2.

Another important consideration is cost. For solutions that address core pain points in the U.S. healthcare system, innovative and flexible payment and reimbursement models may play a key role in supporting sustainable adoption and scaling of beneficial AI technologies. While many clinical AI tools have the potential to reduce total cost of care in aggregate—by preventing downstream complications, shortening length of stay, and reducing avoidable utilization—their value is often realized across different stakeholders than those responsible for initial procurement and implementation. Federal leadership can help bridge this misalignment by advancing payment models that recognize the longitudinal value of diagnostic accuracy, including transitional reimbursement pathways, value-based incentives tied to measurable patient outcomes, and mechanisms that support adoption in resource-constrained hospitals. Aligning reimbursement with demonstrated clinical and economic impact would accelerate responsible AI deployment, ensure equitable access across care settings, and enable sustained innovation while maintaining fiscal stewardship.

HHS question 10: Are there AI research areas HHS should prioritize to accelerate adoption?

a. Are there published findings about the impact of adopted AI tools and their use in clinical care?

There are over 230+ peer-reviewed publications and scientific conference presentations from health systems evaluating the real-world impact of adopted Aidoc solutions in clinical care.

These studies demonstrate how AI can be integrated into radiology triage, care coordination, and patient management workflows in order to reduce time to diagnosis and treatment, decrease emergency department, inpatient, and ICU length of stay, improve awareness of clinically significant findings, increase appropriate and guideline-directed interventions, and reduce patient leakage and gaps in follow-up care.

While these findings are encouraging, important gaps remain in measuring AI's impact on diagnostic quality and patient outcomes. HHS could prioritize research that evaluates how

AI affects diagnostic accuracy, timeliness, and clinical outcomes in real-world care. Although AI has the potential to usher in a new era of diagnostic quality, standardized metrics and validated frameworks to quantify these improvements are still lacking. Federal support for developing these measures will be critical to demonstrating clinical value, building trust, and accelerating responsible adoption. A representative selection of published studies is included as an Appendix.

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

The literature generally evaluates the costs and benefits associated with clinical AI using budget impact models using retrospective economic analyses or interventional (pre/post) studies.

These approaches typically assess financial impact at the patient encounter level and across downstream services, rather than over lifetime horizons. This reflects the relatively early stage of adoption and the limited availability of long-term outcomes and cost data.

Most studies therefore focus on near-term operational and financial effects, such as changes in length of stay, procedural utilization, and follow-up adherence, which can be directly linked to hospital costs and revenues.

Sincerely,

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Appendix: Supporting Literature in response to 10a

Service Line	Module	Metric	Citation Value	Unit	url	Citation
VTE	IVC (Inferior Vena Cava Filter)	Increased Appropriate Interventions	176	%	url	ACC2024 - Randazzo, Michael, et al. "Hospital-Wide Integration of a Natural Language Processing Algorithm to Detect Inferior Vena Cava Filters in Imaging Reports and Improve Device Removal Rates." ACC Abstracts, 2024
VTE	PERT (Pulmonary Embolism Response Team)	Increased Appropriate Interventions	81	%	url	Sanni, Habeeb, et al. "Quantifying the Rise: Artificial Intelligence's Influence on the Volume of Interventional Therapies for Acute Pulmonary Embolism." Pulmonary Artery Interventions (Including PE and PH), Abstract #29459, OR11-1, Society for Cardiovascular Angiography & Interventions, 2024.
VTE	PERT (Pulmonary Embolism Response Team)	Increased Appropriate Interventions	74	%	url	Yun, Jung, et al. "Perfecting PERT: AI-Enhanced Clinical Decision-Making in Acute Pulmonary Embolism Interventions at a Large Academic Center." SIR Abstracts, 2024.
VTE	PERT (Pulmonary Embolism Response Team)	Increased Appropriate Interventions	60	%	url	Zaidan, Mohammed F. "Enhancing Patient Outcomes with an AI-Based PERT in a Large Academic Health System." Abstract presented at the PERT Consortium Annual Symposium, 2023, The University of Texas Medical Branch, Galveston, Texas.
VTE	PERT (Pulmonary Embolism Response Team)	Increased Appropriate Interventions	89.4	%	url	Rali, Parth, et al. "Impact of an AI-Driven Pulmonary Embolism Response Team Workflow on Procedural Volume and Timeliness." Temple University Hospital and Nazareth Hospital, Philadelphia, PA, United States.
Aorta	AA (Aortic Aneurysm)	Reduced Patient Leakage	82	%	url	Kostiuk, Valentyna, et al. "Artificial Intelligence Improves Initial Evaluation and Follow-Up of Patients with AAA." Abstract, Vascular Annual Meeting, 2024. Yale School of Medicine, New Haven, CT.

VTE	IVC (Inferior Vena Cava Filter)	Reduced Patient Leakage	87.2	%	url	Identification of inferior vena cava (IVC) filters on CT, X-ray and MRI radiological reports with a Natural Language Processing (NLP) based tool for management of clinical follow up. RSNA2023. Dr Mojibian
VTE	IVC (Inferior Vena Cava Filter)	Reduced Patient Leakage	79	%	url	Randazzo, Michael, et al. "Hospital-Wide Integration of a Natural Language Processing Algorithm to Detect Inferior Vena Cava Filters in Imaging Reports and Improve Device Removal Rates." ACC Abstracts, 2024.
Radiology	ICH (Intracranial Hemorrhage)	Reduced LoS - ED	36	minutes	url	Davis MA, Rao B, Cedeno PA, Saha A, Zohrabian VM. Machine Learning and Improved Quality Metrics in Acute Intracranial Hemorrhage by Noncontrast Computed Tomography. Curr Probl Diagn Radiol. 2022;51(4):556-561. doi:10.1067/j.cpradiol.2020.10.007
VTE	PERT (Pulmonary Embolism Response Team)	Reduced LoS - ICU	3	days	url	Gupta, Kavish, MD, et al. "Mechanical Thrombectomy, Artificial Intelligence and the Activation of a Pulmonary Embolism Response Team." PERT Consortium 2023, abstract presentation.
VTE	PERT (Pulmonary Embolism Response Team)	Reduced LoS - ICU	2	days	url	Improving Patient Outcomes with an AI-Enhanced Pulmonary Embolism Response Team in a Large Healthcare Network - Burch et al. 2024 PERT Consortium
Radiology	ICH (Intracranial Hemorrhage)	Reduced LoS - IP	2.3	days	url	Davis, Melissa A., et al. "Machine Learning and Improved Quality Metrics in Acute Intracranial Hemorrhage by Noncontrast Computed Tomography." Current Problems in Diagnostic Radiology, vol. 51, no. 4, July–August 2022, pp. 556-561, Elsevier, https://doi.org/10.1067/j.cpradiol.2020.10.007 .
Radiology	ICH (Intracranial Hemorrhage)	Reduced LoS - IP	1.3	days	url	Petry, Michael et al. "Decreased Hospital Length of Stay for ICH and PE after Adoption of an Artificial Intelligence-Augmented Radiological Worklist Triage System." Radiology research and practice vol. 2022 2141839. 18 Aug. 2022, doi:10.1155/2022/2141839
Radiology	PE (Pulmonary Embolism)	Reduced LoS - IP	2.08	days	url	Petry, Michael et al. "Decreased Hospital Length of Stay for ICH and PE after Adoption of an Artificial Intelligence-Augmented Radiological Worklist Triage System." Radiology research and practice vol. 2022 2141839. 18 Aug. 2022, doi:10.1155/2022/2141839

Aorta	AA (Aortic Aneurysm)	Decreased Time to treatment	78	%	url	VAM2024 - Kostiuk, Valentyna, et al. "Artificial Intelligence Improves Initial Evaluation and Follow-Up of Patients with AAA." Abstract, Vascular Annual Meeting, 2024. Yale School of Medicine, New Haven, CT.
VTE	iPE (Incidental Pulmonary Embolism)	Decreased Time to treatment	90	%	url	Radiology AI - Wiklund, Peder, and Koshiar Medson. "Use of a Deep Learning Algorithm for Detection and Triage of Cancer-associated Incidental Pulmonary Embolism." Proceedings of the Radiological Society of North America (RSNA) 2023 Annual Meeting, 18 Oct. 2023, https://doi.org/10.1148/ryai.220286 .
VTE	PE (Pulmonary Embolism)	Decreased Time to treatment	40	%	url	Chalian, et al. "Enhancing Pulmonary Embolism Management: The Impact of AI-Based Triage and Prioritization on Diagnostic Timeliness and Treatment Initiation." RSNA Annual Meeting, 2024.
Neuro	STROKE	Decreased Time to treatment	34	%	url	Chokhawala, H., H. Cuellar, R. Shah, and A. Savardekar. "Enhancing Stroke Care Efficiency through AI Vendor Transition: A Comparative Study of Workflow Metrics Before and After Implementation." SNIS Conference, 2024, LSUHS, Shreveport, LA. Abstract presentation.
Radiology	CSPINE (Cervical Spine Fracture)	Decreased Wait Time	56	%	url	Arash, Dr., et al. "Optimizing Outpatient Care Efficiency: Assessing the Impact of an AI-Powered Triage and Prioritization System on Cervical Spine Fracture Detection through CT Imaging." RSNA Annual Meeting, 2024.
Radiology	Endotracheal Tube (ETT)	Decreased Wait Time	49	%	url	Wismueller "Assessing the Impact of AI-Assisted Triage on Endotracheal Tube Management: A Comparative Analysis of Wait Times." RSNA, 2024.
Radiology	ICH (Intracranial Hemorrhage)	Decreased Wait Time	90	%	url	Ginat, Daniel. "Implementation of Machine Learning Software on the Radiology Worklist Decreases Scan View Delay for the Detection of Intracranial Hemorrhage on CT." Brain Sciences, vol. 11, no. 7, 2021, article 832, doi:10.3390/brainsci11070832.
Radiology	iPE (Incidental Pulmonary Embolism)	Decreased Wait Time	98	%	url	Topff, Laurens, et al. "Artificial Intelligence Tool for Detection and Worklist Prioritization Reduces Time to Diagnosis of Incidental Pulmonary Embolism at CT." RSNA 2023 Scientific Assembly and Annual Meeting Abstracts, 2023.
Radiology	iPE (Incidental Pulmonary Embolism)	Decreased Wait Time	90	%	url	Klein, Dr., et al. "Assessing the Impact of AI-Based Prioritization and Triage of Positive Non-CTPAs on Wait Time in the Outpatient Setting." ATS 2023 Conference, 2023. Poster Presentation.

Radiology	iPE (Incidental Pulmonary Embolism)	Decreased Wait Time	62	%	url	Kahn, Charles E., et al. "Structured Reporting and Communication of Imaging Findings Using Artificial Intelligence: A Consensus Statement of the Society for Imaging Informatics in Medicine." Radiology: Artificial Intelligence, vol. 4, no. 6, 2022, p. e220286, https://doi.org/10.1148/ryai.220286 .
Radiology	PE (Pulmonary Embolism)	Decreased Wait Time	41	%	url	Harrison, Brendan. "Worklist Reprioritization Using Artificial Intelligence Improves Turnaround Times of Chest CT Examinations Positive for Acute Pulmonary Embolism." RSNA Annual Meeting, 2021. Session ID: SDP-MS-2.
VTE	DVT (Deep Vein Thrombosis)	Reduced Readmissions	90	%	url	Outcomes From Mechanical Thrombectomy for Deep Vein Thrombosis: Insights From the PINC AI Healthcare Database Mittleider, Derek et al. Journal of the Society for Cardiovascular Angiography & Interventions, Volume 3, Issue 8, 102149 Secemsky, Eric A et al. "High Burden of 30-Day Readmissions After Acute Venous Thromboembolism in the United States." Journal of the American Heart Association vol. 7,13 e009047. 26 Jun. 2018, doi:10.1161/JAHA.118.009047
VTE	IVC (Inferior Vena Cava Filter)	Reduced Readmissions	62	%	url	Andreoli JM, Thornburg BG, Hickey RM. Inferior Vena Cava Filter-Related Thrombus/Deep Vein Thrombosis: Data and Management. Semin Intervent Radiol. 2016 Jun;33(2):101-4. doi: 10.1055/s-0036-1581087. PMID: 27247478; PMCID: PMC4862844.
VTE	PERT (Pulmonary Embolism Response Team)	Reduced Readmissions	41	%	url	Noman, Anas et al. "Episode Care Costs Following Catheter-Directed Reperfusion Therapies for Pulmonary Embolism: A Literature-Based Comparative Cohort Analysis." The American journal of cardiology vol. 225 (2024): 178-189. doi:10.1016/j.amjcard.2024.06.002